Name ____ Date ___________ Period _____

Parent Functions IV: b^x and log base b - Practice

Algebra II Honors | Unit 1 | Lesson 7 Practice | TEK A2.2.A

Objective: I can convert between exponential and logarithmic form, and reason about logarithm properties.

Section 1 — Recall

1. Convert $4^x = 64$ to logarithmic form, then solve.

Log form: ___________________________ x = __________

2. Convert log base 7 of 49 = 2 to exponential form.

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Section 2 — Apply

3. Solve by inspection: $5^x = 125$, and log base 3 of 27.

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4. Rewrite log(10000) in exponential form and solve.

$10^x = 10000$ x = __________

Section 3 — Challenge

5. Explain why log base b of 1 = 0 for every valid base b (b > 0, b ≠ 1), using the definition of a logarithm as an exponent.

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6. A student solves $2^x = 20$ and gets $x \approx 4.32$ using a calculator. Explain in general terms why an equation like this usually does not have a "nice" whole-number solution, unlike $2^x = 32$.

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Section 4 — Explain It

7. A classmate argues that log base 2 of 8 and log base 8 of 2 must be equal, "since they use the same two numbers, just swapped." Is the classmate correct? Compute both and explain the general pattern.

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